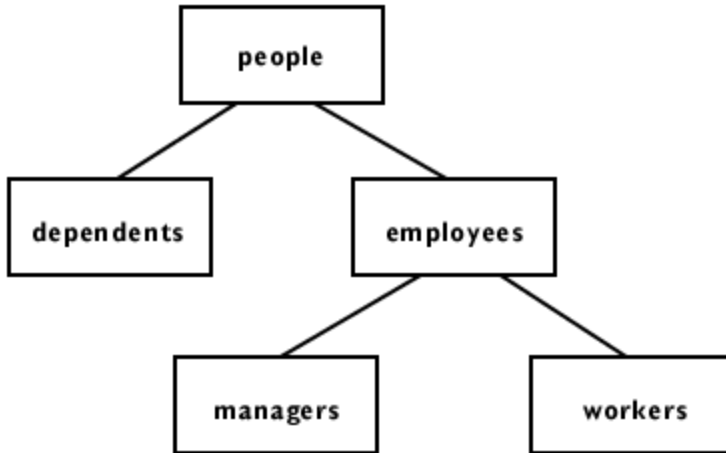


Topic 6.2

6.2: Hierarchies

The world is full of hierarchies. Within a company, we might have the following hierarchy:



There is some information common to *all people*: name, Social Security Number (SSN), date of birth, etc. There is additional information associated with each *employee*: office location, phone number, pay rate, hire date. There is additional information associated with *managers*, not necessarily common to workers, such as departments/projects managed, etc. Similarly, workers have information not necessarily common with managers. The *dependents of an employee*, likewise, have information that employees don't, such as SSN of the employee the dependent is associated with.

The basic idea is that objects at a given level have all the properties of the class above them, plus some new properties unique to their own class. In C++, we use *derived classes* to implement such hierarchies.

Topic 6.3 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-6-dot-3>)